# Notes: Adams, Almeida, and Ferreira (RFS, 2005)

## Powerful CEOs and Their Impact on Corporate Performance

# Contents

# 1 Big picture

- **Research question.** Do CEOs matter more when they have more decision-making power? More specifically, does firm performance become more variable when decision-making power is centralized in the hands of the CEO?
- **Main idea.** If a CEO has strong influence over major decisions, the firm's outcomes reflect that CEO's judgment more directly. This can create more extreme outcomes: very good decisions and very bad decisions are both more likely when fewer other executives constrain or average the CEO's views.
- **Key prediction.** CEO power should increase the *variability* of firm performance. The paper does not primarily predict that powerful CEOs raise or reduce average performance. The main prediction is about dispersion and volatility.
- **Empirical object.** The paper studies 336 nonregulated firms from the 1998 Fortune 500 over 1992–1999. It measures performance using monthly stock returns, return on assets (ROA), and Tobin's $Q$.
- **Main measures of CEO power.** The three central power proxies are whether the CEO is a founder, whether the CEO is the only insider on the board, and whether the CEO has a high concentration of formal titles.
- **Main finding.** Firms with more powerful CEOs have more variable performance. The result is strongest and most consistent for founder CEOs, but the other power measures matter as well, especially in high-managerial-discretion industries.
- **Interpretation.** The result is not necessarily an agency story. Even benevolent CEOs can make mistakes or disagree with other executives. The mechanism is that centralized decision-making reduces diversification of judgment.
- **Economic takeaway.** Powerful CEOs are associated with both the best and the worst outcomes. Weakening CEO power may reduce downside risk, but it may also reduce the chance of extraordinary upside performance.

# 2 Incremental contribution of the paper

- **Moves beyond average performance.** Much governance research asks whether CEO power destroys or creates value on average. This paper argues that the natural implication of CEO influence is higher performance variability, not necessarily a lower mean.
- **Connects corporate finance with organizational theory.** The paper imports ideas from Sah and Stiglitz on group decision-making and from the managerial-discretion literature into an empirical corporate governance setting.
- **Uses heteroskedasticity as the testable implication.** Rather than treating residual variance as a nuisance, the paper makes variance the object of interest and uses heteroskedasticity tests as the main empirical tool.
- **Separates CEO characteristics from organizational structure.** The paper shows that whether a CEO characteristic matters depends on whether the CEO has enough power for that characteristic to influence decisions.

- **Develops structural-power proxies.** The paper focuses on observable institutional features of the executive office: founder status, sole-insider status, and concentration of formal titles.
- **Studies industry discretion.** The paper asks whether CEO power matters more when external constraints are weaker, using Hambrick and Abrahamson's industry managerial-discretion ratings.
- **Clarifies policy implications.** The paper cautions that governance recommendations should not be based only on spectacular failures, because the same concentration of power is also associated with spectacular successes.

# 3 Theory and hypotheses

## 3.1 Definition of power

The paper views powerful CEOs as CEOs who can consistently influence key decisions in their firms despite possible opposition from other executives. This is a decision-rights concept rather than simply a compensation, ownership, or prestige concept.

The paper mainly studies **structural power**: power arising from formal position and titles, founder status, and the CEO's position relative to the board and other top executives.

## 3.2 Decision-making logic

The central theoretical logic is simple:

- Top executives often face ambiguous strategic decisions.
- Different executives may have different judgments, experiences, and biases.
- Group decision-making averages or compromises across opinions.
- CEO-dominated decision-making reflects the CEO's opinion more directly.
- Therefore, CEO-dominated firms should take more extreme decisions.

The paper builds on Sah and Stiglitz's idea that group decision-making diversifies judgment errors. Larger or more influential decision groups are less likely to accept bad projects, but may also be less likely to accept very good projects. A powerful CEO weakens that diversification of judgment.

## 3.3 Main hypothesis

The main empirical hypothesis is:

$$\mathrm{Var}(\text{Firm performance}_{it} \mid \text{CEO power high}) > \mathrm{Var}(\text{Firm performance}_{it} \mid \text{CEO power low}).$$

The hypothesis is about variance, not necessarily mean performance. Powerful CEOs can generate both very good and very bad outcomes.

### 3.4 Why this is not only an agency theory

Agency theory often predicts that powerful CEOs may extract private benefits or evade monitoring. This paper's main mechanism is different. Even when managers have the right incentives, they may make judgment errors. If decisions are centralized, those errors are less diversified.

This distinction matters for interpretation:

- A finding that powerful CEOs have more volatile outcomes does not automatically imply that powerful CEOs are harming shareholders.
- The same mechanism that increases bad outcomes can also increase exceptional good outcomes.
- Governance structures involve a tradeoff between moderation and the possibility of extreme performance.

## 4 Data and measurement

### 4.1 Sample

The sample consists of publicly traded firms in the 1998 Fortune 500 observed over 1992–1999.

- Financial firms and utilities are excluded.
- The final sample contains **336 firms** and **2,633 firm-year observations**.
- Executive and compensation information comes mainly from ExecuComp.
- Monthly stock returns and value-weighted market returns come from CRSP.
- Accounting and segment variables come from Compustat.
- Firm incorporation dates and founder information are collected from Moody's Industrial Manuals, proxy statements, annual reports, and internet sources.

The sample is large-firm focused. The authors note that this may reduce variation in CEO power and performance volatility, but should not mechanically create a positive power-volatility relation.

### 4.2 CEO power measures

The paper uses three main measures.

**1. CEO = founder.** This dummy equals one when the current CEO is one of the firm's founders. Founder CEOs are expected to have more informal authority, status, and influence over the firm.

**2. CEO only insider.** This dummy equals one when no named executive other than the CEO sits on the board. The interpretation is that another inside director could participate in top-level decisions and act as a rival or counterweight to the CEO.

**3. CEO's concentration of titles.** This dummy equals one if the CEO is both chairman and president, or if the CEO is chairman and the firm has no president or COO among the named executives. This captures concentration of formal authority and possible absence of an obvious heir apparent or co-decision-maker.

### 4.3 Performance measures

The paper studies three performance outcomes:

- monthly stock returns, including dividends;
- return on assets, $ROA$ = net income before extraordinary items and discontinued operations/book assets;
- Tobin's $Q$ = (book assets − book equity + market equity)/book assets.

Stock returns are the main performance measure because they capture market valuation changes at a monthly frequency. ROA captures accounting performance. Tobin's $Q$ captures market valuation relative to book assets.

### 4.4 Controls

The main regressions control for CEO ownership, CEO ownership squared, CEO tenure, CEO tenure squared, leverage, firm size, firm age, number of segments, capital expenditures to sales, and industry effects when possible.

These controls matter because firm performance variability may mechanically differ by firm scale, leverage, age, diversification, investment intensity, and CEO ownership.

### 4.5 Managerial discretion

The paper uses Hambrick and Abrahamson's industry ratings of managerial discretion. Industries are classified as high-discretion if they fall in the top 40 percent of the two-digit SIC distribution of the rating and low-discretion if they fall in the bottom 40 percent. Middle industries are dropped to reduce classification noise.

The idea is that CEO power should matter more in industries where managers face fewer environmental constraints and strategic decisions are less mechanically determined by industry conditions.

### 4.6 Alternative power measure: director selection

The paper also tests a board-based measure from Shivdasani and Yermack. The CEO is considered involved in director selection if the firm has no nominating committee, or if the CEO sits on the nominating committee.

This variable captures board influence, but the authors argue it may be less directly tied to day-to-day top-management decision-making than the three main measures.

## 5 Methodology and empirical specifications

### 5.1 Performance residual models

For stock returns, the paper first estimates a market model for each firm:

$$SR_{it} = \beta_i MR_t + u_{it},$$

where $SR_{it}$ is the firm's monthly stock return and $MR_t$ is the CRSP value-weighted market return.

For Tobin's $Q$ and ROA, the paper estimates performance-level models with CEO power variables and controls. The purpose is to remove predictable differences in performance levels before testing whether residual variability differs with CEO power.

### 5.2 Glejser heteroskedasticity test

The main panel test is a Glejser-style regression. After estimating performance residuals, the paper regresses the absolute value of the residual on CEO power and controls:

$$|\hat{u}_{it}| = \alpha + \delta_1 Founder_{it} + \delta_2 OnlyInsider_{it} + \delta_3 Titles_{it} + \Gamma' X_{it} + \varepsilon_{it}.$$

A positive coefficient on a CEO power variable means that performance residuals are larger in absolute value when that form of CEO power is present. This is direct evidence of higher conditional variability.

### 5.3 Within-firm over-time variability

To isolate over-time volatility within each firm, the paper computes the standard deviation of each performance measure over 1992–1999 and estimates:

$$\sigma_i(Performance) = \alpha + \boldsymbol{\delta}' \overline{Power}_i + \Gamma' \overline{X}_i + \text{industry effects} + \varepsilon_i.$$

This test is important because the panel heteroskedasticity tests combine cross-firm and within-firm variation. The standard-deviation regressions ask whether the same firm is more volatile over time when its CEO is more powerful on average.

### 5.4 Industry discretion interactions

The paper then tests whether CEO power matters more in high-discretion industries:

$$Variability_{it} = \alpha + \theta HighDisc_i + \boldsymbol{\delta}' Power_{it} + \boldsymbol{\phi}'(Power_{it} \times HighDisc_i) + \Gamma' X_{it} + \varepsilon_{it}.$$

The interaction coefficients $\boldsymbol{\phi}$ are the key terms. They measure whether CEO power has a stronger effect on performance variability when managers face fewer industry constraints.

### 5.5 Director-selection test

The paper repeats the stock-return variability tests using a dummy for CEO involvement in director selection. It reports both specifications with only the director-selection dummy and specifications that also include the three main CEO power variables.

### 5.6 Endogeneity checks

The paper addresses reverse causality in two ways.

**1. Timing tests.** It checks whether lagged extreme performance predicts increases in CEO power and whether changes in CEO power predict future extreme performance. Extreme performance is defined as very high or very low stock returns.

**2. Instrumental variables.** The paper instruments founder-CEO status using two variables: whether the founder died before the sample period and the number of founders of the firm. The logic is that dead founders cannot be CEOs, while the number of founders mechanically affects the probability that the current CEO is one of them.

The 2SLS estimates for founder CEOs remain positive and significant and are larger than the OLS estimates in the standard-deviation regressions.

# 6 Identification and interpretation

## 6.1 What the paper can identify

The paper identifies a robust association between CEO power and performance variability. The evidence is strongest for founder CEOs and for stock-return volatility. The IV and timing tests support a causal interpretation, but the setting remains observational.

## 6.2 Why variance is the right outcome

The theory predicts that CEO power changes the distribution of outcomes. A powerful CEO should make outcomes more extreme, not necessarily better or worse on average. Therefore, a standard regression of performance levels on CEO power would miss the paper's central implication.

## 6.3 Why the result is not just firm risk

One concern is that risky firms might choose powerful CEOs. The paper reduces this concern in several ways:

- it controls for observable determinants of volatility such as firm size, age, leverage, and diversification;
- it studies over-time variability within firms;
- it tests whether past extreme performance predicts increases in CEO power;
- it instruments founder status using founder death and the number of founders.

## 6.4 Interpretation of founder status

Founder status is the most robust power measure. It likely captures both formal and informal authority: founders may have legitimacy, firm-specific knowledge, loyalty from employees, and reputational status that allows them to override opposition.

However, founder status may also proxy for other firm traits, such as organizational culture or entrepreneurial strategy. The IV tests are meant to reduce this concern, but they do not completely remove all interpretation issues.

### 6.5 Why director selection does not matter

The null result for CEO involvement in director selection suggests that board-level power may not be the relevant source of decision power for performance variability. The top-management team may matter more for operating and strategic decisions than the nominating committee does.

### 6.6 Limitations

- The sample is limited to large Fortune 500 firms and to the 1992–1999 period.
- CEO power is difficult to measure; the proxies capture mainly structural power.
- The paper does not observe the actual decisions made by CEOs or top-management teams.
- Higher variability is not itself a welfare measure. It can reflect both valuable boldness and costly mistakes.

## 7 Tables and figures

### 7.1 Table 1 and Figure 1: Sample characteristics and basic visual evidence

**Read Table 1:**

- Table 1 reports the financial variables, CEO characteristics, and CEO power measures used in the analysis.
- The final sample contains 336 firms and 2,633 firm-year observations.
- Only 9 percent of firm-years have a founder CEO.
- The CEO is the only insider on the board in 29 percent of firm-years.
- The CEO has concentrated titles in 41 percent of firm-years.
- The average firm is large: mean assets are 11.08 billion dollars, reflecting the Fortune 500 sample.

**Read Figure 1:**

- Figure 1 plots average excess stock-return residuals for firms in the low and high quintiles of an aggregate CEO power index.
- The high-CEO-power group contains both the strongest and weakest performers.
- The low-CEO-power group has a visibly narrower distribution.
- The figure provides the paper's central intuition in one picture: power is associated with more extreme outcomes.

**Economic message:** Powerful CEOs are not simply associated with poor performance. They are associated with a wider performance distribution.

**Table 1**
**Summary of select financial variables, CEO characteristics, and power measures**

| Variable | Number of observations | Mean | Standard deviation | Minimum | Maximum |
|---|---|---|---|---|---|
| **Financial variables and CEO characteristics** | | | | | |
| Tobin's *Q* | 2595 | 2.01 | 1.38 | 0.81 | 19.15 |
| Return on assets (*ROA*) | 2633 | 5.42 | 5.80 | −48.19 | 48.15 |
| Stock returns | 30,689 | 0.02 | 0.10 | −0.82 | 2.93 |
| Value-weighted market returns | 30,742 | 0.02 | 0.04 | −0.16 | 0.08 |
| Leverage | 2598 | 0.21 | 0.13 | 0.00 | 1.04 |
| Assets in billions | 2633 | 11.08 | 26.23 | 0.02 | 405.20 |
| Capital expenditures/Sales | 2556 | 0.07 | 0.06 | 0.00 | 1.28 |
| Number of segments | 2543 | 2.74 | 1.70 | 1.00 | 13.00 |
| Firm age | 2622 | 55.43 | 34.53 | 0.00 | 147.00 |
| CEO tenure as CEO | 2257 | 7.37 | 7.22 | 0.00 | 47.00 |
| CEO ownership | 2304 | 0.02 | 0.05 | 0.00 | 0.46 |
| CEO = chair dummy | 2349 | 0.86 | 0.35 | 0.00 | 1.00 |
| CEO = president dummy | 2349 | 0.27 | 0.44 | 0.00 | 1.00 |
| Firm-years in which firm has a president | 2571 | 0.75 | 0.43 | 0.00 | 1.00 |
| Firm-years in which firm has a COO | 2571 | 0.36 | 0.48 | 0.00 | 1.00 |
| **CEO power measures** | | | | | |
| CEO = founder | 2412 | 0.09 | 0.29 | 0.00 | 1.00 |
| CEO only insider | 2571 | 0.29 | 0.45 | 0.00 | 1.00 |
| CEO's concentration of titles | 2349 | 0.41 | 0.49 | 0.00 | 1.00 |

Sample consists of monthly stock returns and all 2633 firm-years of data for 336 publicly traded, non-regulated firms from the 1998 Fortune 500 which were available on ExecuComp (2000) during the years 1992–1999. Data on titles are constructed from 16,022 executive-years of data for these 336 firms during the years 1992–1999. Most financial data, all title data and CEO tenure are from ExecuComp (2000). Monthly stock return data (variable name = RET) and market returns (variable name = VWRETD) are from CRSP. Remaining financial data and segment data are from Compustat. Firm age is collected from Moody's Manuals (1999), proxy statements, and 10-Ks for fiscal 1998. Founder data are collected from a variety of sources consisting of proxy statements, annual reports, and the internet. The number of observations varies because of missing data. Tobin's *Q* = (book value of assets – book value of equity + market value of equity)/book value of assets. *ROA* = net income before extraordinary items and discontinued operations/book value of assets. Leverage = long-term debt/assets. Firm age = number of years since first date of incorporation. Number of segments is equal to the number of different two-digit SIC code industries in which the firm operates. CEO tenure is the number of years since the CEO was appointed CEO. CEO ownership is defined as the ratio of the number of shares owned by the CEO after adjusting for stock splits to total shares outstanding. CEO = founder is equal to one if the CEO is a founder of the company. CEO only insider is equal to one if the CEO is the only insider on the board of directors. CEO's concentration of titles is a dummy that is equal to one when both the CEO = chair and the CEO = president dummies are equal to one, or when the CEO = chair dummy is equal to one and the firm has neither a president nor a COO in that year.

Table 1: Summary statistics for financial variables, CEO characteristics, and CEO power measures.

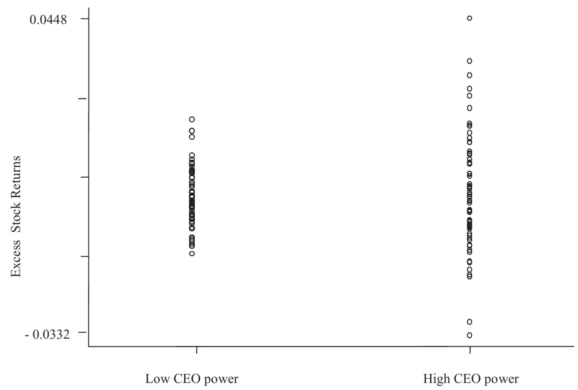


**Figure 1**
**Best and worst performances and CEO power**
Figure 1 shows a plot of average stock-return residuals for very low and very high values of CEO power. The average excess stock return is the difference between a firm's average stock return and its beta times the average market return, where firm betas are estimated from a standard market model using monthly stock returns over the 1992–1999 period. Average returns are averages of monthly returns over the period 1992–1999. We construct an aggregate power index that is the sum of our three measures of CEO power, CEO = founder, CEO only insider, and CEO's concentration of titles. We categorize firms' CEOs as having high power if the average of their aggregate power index is in the fifth quintile. We categorize firms' CEOs as having low influence  power if the average of their aggregate power index is in the first quintile.

Figure 1: Best and worst performances by CEO power group.

## 7.2 Tables 2 and 3: Main performance-variability tests

**Read Table 2:**

- Table 2 reports Glejser heteroskedasticity tests using absolute performance residuals.
- In the stock-return regression, founder CEOs and CEO-only-insider status are positive and statistically significant.
- Founder status is also positive and significant for the variability of ROA residuals and Tobin's $Q$ residuals.
- The three CEO power variables are jointly significant for all three performance measures.
- The economic magnitudes are large: moving from nonfounder to founder CEO increases the absolute value of stock-return residuals by roughly 18.5 percent relative to the sample average reported by the authors.

**Read Table 3:**

- Table 3 uses each firm's standard deviation of performance over 1992–1999 as the dependent variable.
- Founder status significantly increases the standard deviation of stock returns and Tobin's $Q$.
- In the ROA specification, all three power measures are positive and statistically significant or marginally significant.

- The three power variables are jointly significant across the specifications.
- This test helps show that the result is not only a cross-sectional comparison across firms.

**Economic message:** CEO power is associated with both cross-firm and within-firm-over-time performance variability. Founder status is the most robust proxy.

**Table 2**
**Heteroskedasticity tests for performance measures as a function of CEO power and other control variables**

| Dependent variable | Absolute value of excess stock returns (I) | Absolute value of *ROA* residuals (II) | Absolute value of Tobin's $Q$ residuals (III) |
|---|---|---|---|
| CEO = founder | 0.012*** | 0.985** | 0.416*** |
| | (4.49) | (2.19) | (4.28) |
| CEO only insider | 0.003** | −0.218 | 0.032 |
| | (2.24) | (−1.09) | (0.68) |
| CEO's concentration of titles | 0.001 | 0.303 | 0.027 |
| | (0.82) | (1.55) | (0.60) |
| CEO ownership | 0.158*** | 7.945 | 6.250*** |
| | (5.03) | (1.44) | (3.08) |
| CEO ownership squared | −0.378*** | −11.470 | −18.660*** |
| | (−4.06) | (−0.59) | (−3.02) |
| CEO tenure | −0.001 | −0.375 | −0.102* |
| | (−0.81) | (−1.19) | (−1.90) |
| CEO tenure squared | −0.001 | 0.042 | 0.022 |
| | (−1.63) | (0.44) | (1.19) |
| Leverage | 0.027*** | −1.489 | −0.822*** |
| | (5.27) | (−1.29) | (−4.25) |
| Firm size | −0.004*** | −0.084 | 0.067*** |
| | (−6.48) | (−0.73) | (2.6) |
| Firm age | −0.001*** | −0.018 | −0.011 |
| | (−6.61) | (−0.54) | (−1.57) |
| Number of segments | −0.001** | −0.281*** | −0.048*** |
| | (−2.02) | (−5.79) | (−4.67) |
| Capex/sales | 0.013 | 1.322 | 0.652* |
| | (1.10) | (0.81) | (1.65) |
| Constant | 0.123*** | 6.104*** | 0.753*** |
| | (21.66) | (5.63) | (3.21) |
| Number of observations | 24,540 | 2078 | 1953 |
| *F*-statistic of joint significance test for CEO = founder, CEO only insider, and CEO's concentration of titles | 10.10*** | 2.74** | 6.13*** |
| $R^2$ | 0.11 | 0.11 | 0.22 |

Table 2 shows the results of using Glejser's (1969) method to test whether the variance in performance is greater in firms in which our measures of CEO power are larger. To perform the tests for stock returns we construct excess stock returns $\hat{u}_i$ for firm $i$ from the market model: $SR_{it} = \beta_i MR_t + u_{it}$, where $SR$ denotes monthly stock returns, $MR$ the monthly value-weighted market return, and $t$ ranges from January, 1992 to December, 1999. We construct residuals $\hat{u}_i$ for Tobin's $Q$ from the following regression: $Q = b_0 + b_1$CEO = founder + $b_2$CEO only insider + $b_3$CEO's concentration of titles + $b_4$CEO ownership + $b_5$CEO ownership squared + $b_6$ROA+ $b_7$ one period lagged $ROA$ + $b_8$capex/sales + $b_9$firm age + $b_{10}$ number of segments + $b_{11}$ln (assets) + year dummies. We construct $ROA$ residuals using the following model: $ROA = b_0 + b_1$CEO = founder + $b_2$CEO only insider + $b_3$CEO's concentration of titles + $b_4$CEO ownership + $b_5$CEO ownership squared + $b_6$capex/sales + $b_7$firm age + $b_8$ number of segments + $b_9$ln (assets) + year dummies. We regress the absolute value of the residuals from these regressions on CEO = founder, CEO's concentration of titles and CEO only insider and controls including CEO ownership (measured by the ratio of the number of shares owned by the CEO after adjusting for stock splits to total shares outstanding), squared CEO ownership, CEO tenure (the number of years since the CEO was appointed CEO) and its square, leverage (long-term debt/assets), firm size (natural log of assets), firm age (number of years since first date of incorporation), Capex/sales and number of segments (two-digit SIC segments the firm operates in). All regressions include two-digit SIC code industry dummies. Robust $t$-statistics are in parentheses. ***, **, and * indicate significance at the 1, 5, and 10% levels, respectively. The coefficients on CEO tenure and firm age are multiplied by 10. The coefficient on CEO tenure squared is multiplied by 100.

Table 2: Glejser heteroskedasticity tests for performance residuals.

**Table 3**
**Cross-sectional regressions of standard deviation of performance measures on CEO power and other control variables**

| Dependent variable | Standard deviation of stock returns (I) | Standard deviation of *ROA* (II) | Standard deviation of Tobin's $Q$ (III) |
|---|---|---|---|
| CEO = founder | 0.021*** | 1.337* | 0.616*** |
| | (2.71) | (1.77) | (2.77) |
| CEO only insider | 0.007 | 1.016* | −0.079 |
| | (1.37) | (1.81) | (−0.58) |
| CEO's concentration of titles | 0.005 | 0.863* | 0.105 |
| | (1.16) | (1.92) | (0.92) |
| CEO ownership | 0.205** | 1.145 | 6.998 |
| | (2.01) | (0.11) | (1.44) |
| CEO ownership squared | −0.487 | 13.739 | −16.509 |
| | (−1.65) | (0.49) | (−1.24) |
| CEO tenure | −0.001 | −0.085 | 0.006 |
| | (−0.96) | (−1.28) | (0.50) |
| CEO tenure squared | −0.073 | 17.268 | −6.987 |
| | (−0.38) | (0.84) | (−1.57) |
| Leverage | 0.035*** | −0.451 | −1.515*** |
| | (2.70) | (−0.30) | (−4.47) |
| Firm size | −0.008*** | −0.246 | 0.037 |
| | (−4.95) | (−1.17) | (0.98) |
| Firm age | −0.016*** | 0.344 | −0.107 |
| | (−3.29) | (0.54) | (−0.92) |
| Number of segments | −0.019 | −24.560** | −3.379* |
| | (−0.16) | (−2.21) | (−1.76) |
| Capex/sales | 0.071 | 0.526 | 1.259 |
| | (1.05) | (0.10) | (1.26) |
| Constant | 0.161*** | 5.485*** | 0.443 |
| | (11.34) | (3.06) | (1.22) |
| Number of observations | 320 | 320 | 320 |
| *F*-statistic of joint significance test for CEO = founder, CEO only insider and CEO's concentration of titles | 4.70*** | 3.42** | 2.90** |
| $R^2$ | 0.58 | 0.28 | 0.41 |

Table 3 shows cross-sectional OLS regressions of the standard deviation of performance measures, computed for each firm over the 1992–1999 period, on CEO = founder, CEO's concentration of titles and CEO only insider and controls averaged over the 1992–1999 period. In column I, the dependent variable is the standard deviation of monthly stock returns ($SR$). In columns II and III, the dependent variables are the standard deviations of $ROA$ and Tobin's $Q$. The regressions in all columns include two-digit SIC code dummies. Robust $t$-statistics are shown in parentheses. ***, **, and * indicate significance at the 1, 5, and 10% levels, respectively. The coefficient on CEO tenure squared is multiplied by 10,000. The coefficients on firm age and number of segments are multiplied by 100.

Table 3: Standard-deviation regressions for stock returns, ROA, and Tobin's $Q$.

## 7.3 Tables 4 and 5: Managerial discretion as a moderator

**Read Table 4:**

- Table 4 studies stock-return variability using the high-discretion-industry indicator and its interactions with CEO power.
- High-discretion industries have more volatile stock returns in specifications without interactions.
- Once interactions are added, the direct CEO power effects in low-discretion industries become small.
- The interactions between high discretion and founder status, CEO-only-insider status, and concentrated titles are positive, with strong significance in the Glejser tests.

- This means CEO power matters especially where managers have room to make consequential choices.

**Read Table 5:**

- Table 5 repeats the discretion-interaction tests using ROA and Tobin's $Q$.
- The results are broadly consistent with Table 4: CEO power has stronger positive effects on variability in high-discretion industries.
- The evidence is strongest in the Glejser tests and in the ROA standard-deviation regressions.
- The results support the managerial-discretion view that organizational power matters most when external constraints are weak.

**Economic message:** CEO power is not equally important everywhere. It becomes more consequential in industries where executives have more strategic discretion.

**Table 4**
**Stock return variability as a function of CEO power interacted with industry ratings of managerial discretion**

| Dependent variable | Absolute value of excess stock returns (I) | Absolute value of excess stock returns (II) | Standard deviation of stock returns (III) | Standard deviation of stock returns (IV) |
|---|---|---|---|---|
| High-discretion industry | 0.006*** | −0.001 | 0.012*** | −0.001 |
| | (3.88) | (−0.37) | (3.24) | (−0.28) |
| CEO = founder | 0.017*** | 0.002 | 0.027*** | 0.005 |
| | (5.55) | (0.63) | (3.21) | (0.49) |
| CEO only insider | 0.004** | −0.002 | 0.009 | −0.003 |
| | (2.42) | (−1.40) | (1.42) | (−0.43) |
| CEO's concentration of titles | 0.020 | 0.008 | 0.062 | 0.028 |
| | (1.61) | (0.52) | (1.42) | (0.53) |
| CEO = founder* high-discretion industry | – | 0.018*** | – | 0.029** |
| | | (3.41) | | (2.06) |
| CEO only insider* high-discretion industry | – | 0.013*** | – | 0.023* |
| | | (4.01) | | (1.92) |
| CEO's concentration of titles*high-discretion industry | – | 0.004* | – | 0.012 |
| | | (1.71) | | (1.38) |
| Type of test | Glejser | Glejser | Variability over time | Variability over time |
| Number of observations | 16,094 | 16,094 | 209 | 209 |
| $F$-statistic of joint significance test for the interaction terms | – | 11.66*** | – | 4.25*** |
| $R^2$ | 0.09 | 0.10 | 0.47 | 0.51 |

In Table 4, we examine whether our measures of CEO power have greater impact on the variability of performance when CEOs are in industries where they are likely to have more discretion. We construct our measure of managerial discretion using Hambrick and Abrahamson's (1995) ratings of managerial discretion for seventy four-digit SIC code industries. We average their measures by two-digit SIC code industry, and then we construct a variable which is equal to 1 if the firm belongs to an industry in the top 40% of the distribution of the rating of managerial discretion, and is equal to 0 if the firm belongs to an industry in the bottom 40% of the distribution of the rating of managerial discretion. We examine the direct role of this variable in columns I and III, and we interact it with our measures of CEO power in columns II and IV. Columns I and II show the results using the Glejser test, and columns III and IV show cross-sectional OLS regressions of the standard deviation in stock returns, computed for each firm over the 1992–1999 period, on our measures of power and the same controls we use in Tables 2 and 3 averaged over the 1992–1999 period. The coefficients and $t$-statistics on the controls are omitted for the sake of brevity. The regressions do not include two-digit SIC industry dummies because they are collinear with the industry ratings. Robust $t$-statistics are shown in parentheses. ***, **, and * indicate significance at the 1, 5, and 10% levels. The coefficient on CEO's concentration of titles is multiplied by 10.

Table 4: Stock-return variability and CEO power interacted with industry managerial discretion.

**Table 5**
**Variability in Tobin's *Q* and *ROA* as a function of CEO power interacted with industry ratings of managerial discretion**

| Dependent variable | Absolute value of *ROA* residuals (I) | Absolute value of Tobin's *Q* residuals (II) | Standard deviation of *ROA* (III) | Standard deviation of Tobin's *Q* (IV) |
|---|---|---|---|---|
| High discretion industry | 0.771*** | 0.523*** | −1.020* | 0.152 |
| | (2.77) | (7.64) | (−1.79) | (1.01) |
| CEO = founder | −0.141 | 0.115 | −1.045 | 0.188 |
| | (−0.29) | (0.97) | (−1.17) | (0.89) |
| CEO only insider | −0.563*** | −0.132*** | 0.303 | −0.131 |
| | (−2.72) | (−3.65) | (0.49) | (−1.18) |
| CEO's concentration of titles | −0.152 | −0.049 | 0.040 | 0.050 |
| | (−0.79) | (−1.43) | (0.10) | (0.65) |
| CEO = founder*high-discretion industry | 0.559 | 0.356* | 2.637* | 0.688* |
| | (0.80) | (1.81) | (1.96) | (1.68) |
| CEO only insider*high-discretion industry | 0.817 | 0.241* | 2.697* | −0.031 |
| | (1.47) | (1.64) | (1.77) | (−0.12) |
| CEO's concentration of titles* high-discretion industry | 0.823* | −0.003 | 1.684* | 0.214 |
| | (1.86) | (−0.03) | (1.87) | (0.74) |
| Type of test | Glejser | Glejser | Variability over time | Variability over time |
| Number of observations | 1367 | 1367 | 209 | 209 |
| $F$-statistic of joint significance test for the interaction terms | 2.20* | 1.89 | 4.01*** | 0.95 |
| $R^2$ | 0.09 | 0.24 | 0.24 | 0.31 |

In Table 5, we replicate the tests reported in Table 4 using Tobin's $Q$ and $ROA$ as alternative performance measures. We construct our measure of managerial discretion using Hambrick and Abrahamson's (1995) ratings of managerial discretion, as described in Table 4. Columns I and II show the results of the Glejser tests, where the residuals of $ROA$ and Tobin's $Q$ are constructed as in Table 2. Columns III and IV show the results of cross-sectional OLS regressions of the standard deviation in $ROA$ and Tobin's $Q$, computed for each firm over the 1992–1999 period, on our measures of power and the same controls we use in Tables 2 and 3 averaged over the 1992–1999 period. The coefficients and $t$-statistics on the controls are omitted for the sake of brevity. The regressions do not include two-digit SIC industry dummies because they are collinear with the industry ratings. Robust $t$-statistics are shown in parentheses. ***, **, and * indicate significance at the 1, 5, and 10% levels.

Table 5: ROA and Tobin's $Q$ variability interacted with industry managerial discretion.

## 7.4 Table 6: CEO involvement in director selection

**Read:**

- Table 6 tests whether CEO involvement in director selection predicts stock-return variability.
- The CEO is involved in director selection when the firm has no nominating committee or when the CEO sits on the nominating committee.
- The coefficient on director-selection involvement is not statistically significant.
- The main CEO power variables retain their earlier signs and significance patterns when included.

- The result suggests that board-nomination influence is not the same as operational or executive-office decision power.

**Economic message:** The relevant power for performance variability appears to be power over top-level decision-making, especially inside the executive team, rather than simply involvement in selecting directors.

**Table 6**
**Tests using a variable indicating whether the CEO is involved in the selection of directors**

| Dependent variable | Absolute value of excess stock returns (I) | Absolute value of excess stock returns (II) | Standard deviation of stock returns (III) | Standard deviation of stock returns (IV) |
|---|---|---|---|---|
| CEO involved in director selection | −0.041 (−0.37) | −0.010 (−0.09) | −0.244 (−0.66) | −0.101 (−0.28) |
| CEO = founder | – | 0.013*** (4.67) | – | 0.021*** (2.70) |
| CEO only insider | – | 0.003** (2.20) | – | 0.007 (1.32) |
| CEO's concentration of titles | – | 0.007 (0.63) | – | 0.047 (1.16) |
| CEO ownership | 0.175*** (5.42) | 0.155*** (4.94) | 0.235** (2.26) | 0.207** (2.01) |
| CEO ownership squared | −0.368*** (−3.9) | −0.368*** (−3.97) | −0.458 (−1.53) | −0.490 (−1.64) |
| CEO tenure | −0.001 (−0.8) | −0.001 (−0.73) | −0.001 (−1.04) | −0.007 (−0.97) |
| CEO tenure squared | −0.005 (−0.95) | −0.010* (−1.67) | 0.003 (0.16) | −0.006 (−0.33) |
| Leverage | 0.028*** (5.48) | 0.027*** (5.36) | 0.040*** (2.74) | 0.035*** (2.69) |
| Firm size | −0.004*** (−6.76) | −0.004*** (−6.34) | −0.009*** (−5.03) | −0.009*** (−4.95) |
| Firm age | −0.001*** (−7.86) | −0.001*** (−6.48) | −0.002*** (−4.3) | −0.002*** (−3.34) |
| Number of segments | −0.008** (−2.3) | −0.008** (−2.31) | −0.001 (−0.09) | −0.002 (−0.14) |
| Capex/sales | 0.008 (0.53) | 0.007 (0.44) | 0.077 (1.14) | 0.072 (1.05) |
| Constant | 0.093*** (17.93) | 0.088*** (16.19) | 0.171*** (11.58) | 0.162*** (11.20) |
| Type of test | Glejser | Glejser | Variability over time | Variability over time |
| Number of observations | 24,190 | 24,190 | 320 | 320 |
| $R^2$ | 0.11 | 0.11 | 0.56 | 0.58 |

Columns I and II show the results of using the Glejser test to test whether the variance in performance is greater in firms in which the CEO is involved in the selection of directors. To perform the tests for stock returns, we construct excess stock returns $\hat{u}_i$ for firm $i$ from the market model: $SR_{it} = \beta_i MR_t + u_{it}$, where $SR$ denotes monthly stock returns, $MR$ the monthly value-weighted market return and $t$ ranges from 1992 to 1999. We regress the absolute value of the residuals on a dummy indicating whether the CEO is involved in director selection and controls including CEO = founder, CEO only insider, and CEO's concentration of titles. Following Shivdasani and Yermack (1997), the CEO is defined to be involved in director selection if the firm does not have a nominating committee, or if the firm has a nominating committee and the CEO sits on the committee. Columns III and IV show cross-sectional OLS regressions of the standard deviation in stock returns, computed for each firm over the 1992–1999 period, on the same variables. The regressions in all columns include two-digit SIC code dummies. Robust $t$-statistics are shown in parentheses. ***, **, and * indicate significance at the 1, 5, and 10% levels. The coefficient on CEO involved in director selection is multiplied by 100. The coefficients on CEO's concentration of titles, CEO tenure, firm age and number of segments are multiplied by 10. The coefficient on CEO tenure squared is multiplied by 1000.

Table 6: CEO involvement in director selection and stock-return variability.

# 8 Short summary

- The paper asks whether powerful CEOs have a larger impact on firm outcomes.
- Its key prediction is that CEO power increases *performance variability*, not necessarily average performance.
- The main theoretical mechanism is diversification of judgment: groups moderate individual errors, while powerful CEOs make decisions reflect their own views more directly.

- The sample includes 336 large, nonregulated Fortune 500 firms over 1992–1999.
- CEO power is measured by founder status, sole-insider status on the board, and concentration of CEO titles.
- The paper uses heteroskedasticity tests and standard-deviation regressions to test whether performance is more variable under powerful CEOs.
- Founder CEOs are the most robustly associated with higher variability in stock returns, ROA, and Tobin's $Q$.
- CEO-only-insider status is especially important for stock-return variability.
- CEO power matters more in industries with high managerial discretion.
- CEO involvement in director selection does not explain performance variability.
- Endogeneity checks suggest that the evidence is consistent with causality running from CEO power to performance variability.
- The paper's governance implication is nuanced: reducing CEO power may reduce extreme failures, but may also reduce extreme successes.

# 9 Conclusion

## 9.1 Core conclusion

Adams, Almeida, and Ferreira show that firm performance is more variable when decision-making power is more centralized in the hands of the CEO. The most robust evidence comes from founder CEOs, but the broader pattern is that CEO power becomes more important when the CEO operates in a high-discretion environment.

## 9.2 Economic intuition

The intuition is that group decision-making diversifies judgment. A less powerful CEO must compromise with other executives, which moderates outcomes. A powerful CEO can push through decisions that reflect his own beliefs more fully. Those decisions can be unusually good or unusually bad.

## 9.3 Incremental contribution revisited

The paper reframes the CEO power debate. Rather than asking only whether powerful CEOs are good or bad, it asks whether powerful CEOs matter more. The answer is yes: powerful CEOs are associated with a wider distribution of firm outcomes.

## 9.4 Implications

The findings imply that governance reforms should consider the full distribution of outcomes. Separating CEO and chairman roles, adding inside counterweights, or reducing founder dominance may reduce the risk of extreme negative outcomes. But those changes may also reduce the chance of extreme positive outcomes.

### 9.5 Limitations

The empirical setting is observational, and CEO power is only imperfectly measured. The paper cannot observe the exact decision process inside the top-management team. It also focuses on large firms in the 1990s, so external validity to smaller firms, startups, or more recent governance regimes requires caution.

### 9.6 Takeaway

The enduring takeaway is that CEOs matter most when they have power. Concentrated CEO authority makes firm outcomes more extreme by allowing the CEO's judgment to shape decisions more directly. This is a risk-return tradeoff in organizational design, not simply a story of good or bad governance.